Roll No.: DL.AI.U4AID24140

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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2024

First Semester

Artificial Intelligence

23BIO112 – INTRODUCTION TO BIOLOGICAL DATA

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Demonstrate a clear understanding of the structure and function of cells, the basics of pathology, and molecular biology concepts, including genomics, proteomics, and protein families.
CO02	Identify and describe various types of data & biological data, such as genomic, proteomic, and biomedical signal data, and effectively utilize relevant biological and protein functional databases.
CO03	Apply data preprocessing techniques, including cleaning, normalization, and annotation, and use visualization tools to interpret and present biological data.
CO04	Interpret medical imaging data (e.g., X-ray, CT scans, MRI) and biomedical signals (e.g. EEG), with a particular focus on applications in mental health.

Answer all questions

PART A **[10 * 1 = 10 Marks]**

1. Which structure is primarily responsible for maintaining homeostasis within a living organism?
   A) Organ B) Tissue C) Cell D) System
   [1] [CO01] [BTL 1]
2. What type of nucleic acid is primarily involved in transferring genetic information from DNA to the ribosome for protein synthesis?
   A) mRNA B) tRNA C) rRNA D) DNA
   [1] [CO01] [BTL 1]
3. What is the process of making an RNA copy of a DNA sequence called?
   A) Translation B) Transcription C) Replication D) Duplication
   [1] [CO01] [BTL 1]
4. During which phase of the cell cycle does DNA replication occur?
   A) G1 Phase B) S Phase C) G2 Phase D) M Phase
   [1] [CO01] [BTL 1]
5. Which organelle is responsible for protein synthesis?
   A) Mitochondria B) Ribosome C) Nucleus D) Golgi apparatus
   [1] [CO01] [BTL 1]

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6. What is the primary role of the enzyme DNA polymerase during DNA replication?
   A) Unwind the DNA double helix
   B) Synthesize RNA primers
   C) Add nucleotides to growing DNA strand
   D) Seal gaps between Okazaki fragments
   [1] [CO01] [BTL 1]

7. Which of the following correctly describes the outcome of mitosis?
   A) Two haploid daughter cells
   B) Four haploid daughter cells
   C) Two diploid daughter cells
   D) Four diploid daughter cells
   [1] [CO01] [BTL 1]

8. Which of the following measures demonstrates resilience to the effects of outliers in a dataset?
   A) Mean B) Median C) Mode D) Standard Deviation
   [1] [CO02] [BTL 2]

9. Explain which measure of dispersion is most suitable for representing variability in a dataset with a significant number of outliers, as it is least affected by extreme values.
   A) Standard Deviation B) Range C) Interquartile Range (IQR) D) Variance
   [1] [CO02] [BTL 2]

10. Which organelle is responsible for synthesizing lipids and storing calcium in muscle cells?
    A) Rough Endoplasmic Reticulum
    B) Smooth Endoplasmic Reticulum
    C) Mitochondria
    D) Lysosome
    [1] [CO01] [BTL 1]

PART B **[6 * 5 = 30 Marks]**

11. Make use of single nested VLOOKUP(). Explain the syntax and solve the following:

Sheet 1: Patient

PID	PNAME	DID
P01	AAA	D01
P02	BBB	D02
P03	CCC	D01

Sheet 2: Doctor

DID	DNAME
D01	XXX
D02	YYY

Sheet 3: Consultant_fee

DID	DFEE
D01	1500
D02	2000

Find both the treating Doctor's Name and their Consultant Fee for the Patient ID P03.
[5] [CO02] [BTL 3]

12. A biologist is conducting an experiment to analyze the nucleotide composition of a newly discovered DNA sample extracted from an ancient organism. He has identified the percentage of nucleotide base - Adenine (A) as 38%. Solve using Chargaff's Rule and determine the missing percentages of the remaining nucleotide bases. [5] [CO02] [BTL 3]

13. Explain the central dogma of molecular biology and the key processes involved in synthesizing proteins. [5] [CO01] [BTL 2]

14. Label any 10 organelles in an animal cell and their functions with a neat diagram.
[5] [CO01] [BTL 1]

15. Compare and contrast the features of structured, semi-structured and unstructured data.
[5] [CO02] [BTL 1]

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16. Outline the process of DNA replication and highlight the roles of key enzymes involved.
[5] [CO01] [BTL 2]

PART C **[1 * 10 = 10 Marks]**

17. Explain in detail about the process of meiotic cell division and its importance in sexual reproduction. [10] [CO01] [BTL 2]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**33**	BTL 1	**18**
CO02	**17**	BTL 2	**22**
CO03	-	BTL 3	**10**
CO04	-	BTL 4	
		BTL 5	
		BTL 6	